DIGITAL FORENSICS LABORATORY REPORT

Live Memory Forensics with Volatility 3

Acquisition and Plugin-Based Triage of a Windows Memory Image

DUMPIT · VOLATILITY 3 · WINDOWS MEMORY ANALYSIS

REPORT ID	DFCS · CCFA · VOLATILITY · 2026-07
PREPARED BY	Awais Ahmed
ACADEMIC PROGRAM	B.S. Digital Forensics and Cyber Security (DFCS)
CERTIFICATION TRACK	Certified Computer Forensics Analyst (CCFA)
SUBJECT MEDIA	Full physical memory image of the examiner's own HP laptop (LAPTOP-QRAGDIG9), ≈ 5.9 GB
ACQUISITION TOOL	DumpIt (h4sh5/DumpIt-mirror)
ANALYSIS TOOL	Volatility 3 Framework 2.28.1 (Python 3.12)
REPORT DATE	July 23, 2026
CLASSIFICATION	Academic Coursework: Portfolio Release Copy

This document reproduces coursework performed against a live memory capture of the examiner's own personal system, in a controlled lab environment, for digital forensics education. All steps were independently executed by the author.

Report Details

Field	Detail
Examiner	Awais Ahmed
Role	Digital Forensics Student / Examiner-in-Training
Source Practical	“Volatility” memory-forensics worksheet
Subject Media	Physical memory image of the examiner's own system, captured for this exercise

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1. Executive Summary

This report documents a live memory (RAM) forensics exercise performed as part of Digital Forensics and Cyber Security (DFCS) coursework aligned to Certified Computer Forensics Analyst (CCFA) competencies. A full physical memory image was captured from the examiner's own Windows 10 laptop using DumpIt, then analyzed with the Volatility 3 Framework to demonstrate the standard triage workflow an examiner would follow against any Windows memory capture.

Seven Volatility plugins were run against the image in sequence, windows.info, windows.pslist, windows.netscan, windows.pstree, windows.cmdline, windows.malware.malfind, and windows.filescan, each targeting a different category of volatile evidence: system identification, running processes, active network connections, process ancestry, process launch arguments, injected-code detection, and open file handles.

No malicious activity was identified. windows.malware.malfind flagged three processes with executable-writable memory regions, all of which were assessed as expected false positives tied to legitimate OEM and antivirus software rather than injected code (Section 5.6). The report closes with the command syntax for narrowing a Volatility search around a specific suspicious process, which would be the next step if this had been a live incident-response engagement rather than a clean baseline system.

2. Scope and Authorized Use

All actions in this report were performed against a memory image of the examiner's own personal system, captured specifically for this coursework exercise, in a controlled lab environment. Capturing and analyzing full system memory is a standard, lawful step in an authorized digital forensics or incident-response engagement; performing the same capture against a system one does not own or have written authorization to examine would not be lawful. This report should be read strictly as a record of coursework against the examiner's own machine, not as guidance for use against systems without proper authorization.

3. Background: Why Memory Forensics Matters

Traditional (“dead-box”) forensics, as in the companion Windows Registry examination of this portfolio, recovers evidence from a disk image after the system has been shut down. Memory forensics instead captures the contents of RAM while a system is running, and analyzes that snapshot afterward. This matters because a large amount of forensically valuable information exists only in memory and is lost the moment a system powers off or a process exits:

- Running processes and their exact command-line arguments, including ones that never wrote a trace to disk.
- Active and recently-closed network connections, tied to the specific process that owned them.
- Decrypted data, plaintext credentials, or encryption keys that exist in memory only while a program is actively using them.
- Malicious code injected directly into a legitimate process's memory space, which may never exist as a file on disk at all.

Volatility 3 is the current generation of the Volatility Framework, an open-source tool purpose-built to parse these raw memory images and reconstruct operating-system structures (the process list, network tables, loaded modules, and so on) directly from the raw bytes of a memory dump, without needing the original live system.

4. Examination Environment and Tools

- Python 3.12, required runtime for Volatility 3.
- Volatility 3 Framework 2.28.1, installed in editable mode from source with the full optional dependency set (yara-python, capstone, pycryptodome, leechcorepyc, pefile), enabling malware-scanning and disassembly features used later in this report.

Volatility 3 source: https://github.com/volatilityfoundation/volatility3

- DumpIt (h4sh5/DumpIt-mirror), a lightweight Windows memory-acquisition utility used to capture the full physical memory image analyzed in this report.

DumpIt-mirror source: https://github.com/h4sh5/DumpIt-mirror

5. Procedure and Results

5.1 Installing Volatility 3

Volatility 3 was installed in editable mode with its full optional dependency set, which pulls in pefile, yara-python, capstone, pycryptodome, and leechcorepyc, the packages that specifically enable PE parsing, YARA pattern scanning, disassembly, and low-level memory-layer support used by later plugins in this report.

```
pip install --user -e ".[full]"
```

```
C:\Users\Awais\Desktop\CCFA\3. Tools\volatility3-develop>pip install --user -e ".[full]"
Obtaining file:///C:/Users/Awais/Desktop/CCFA/3.%20Tools/volatility3-develop
  Installing build dependencies ... done
  Checking if build backend supports build_editable ... done
  Getting requirements to build editable ... done
  Preparing editable metadata (pyproject.toml) ... done
Collecting pefile>=2024.8.26 (from volatility3==2.28.1)
  Downloading pefile-2024.8.26-py3-none-any.whl.metadata (1.4 kB)
Collecting yara-python<5,>=4.5.1 (from volatility3==2.28.1)
  Downloading yara_python-4.5.4-cp312-cp312-win_amd64.whl.metadata (3.0 kB)
Collecting capstone<6,>=5.0.3 (from volatility3==2.28.1)
  Downloading capstone-5.0.9-py3-none-win_amd64.whl.metadata (3.4 kB)
Collecting pycryptodome<4,>=3.21.0 (from volatility3==2.28.1)
  Downloading pycryptodome-3.23.0-cp37-abi3-win_amd64.whl.metadata (3.5 kB)
Collecting leechcorepyc<3,>=2.19.2 (from volatility3==2.28.1)
  Downloading leechcorepyc-2.22.12-cp36-abi3-win_amd64.whl.metadata (536 bytes)
Requirement already satisfied: pillow<11.0.0,>=10.0.0 in c:\users\awais\appdata\local\programs\python\python312\lib\s
-packages (from volatility3==2.28.1) (10.3.0)
Downloading capstone-5.0.9-py3-none-win_amd64.whl (1.3 MB)
   ---------------------------------------- 1.3/1.3 MB 250.8 kB/s eta 0:00:00
Downloading leechcorepyc-2.22.12-cp36-abi3-win_amd64.whl (494 kB)
Downloading pefile-2024.8.26-py3-none-any.whl (74 kB)
Downloading pycryptodome-3.23.0-cp37-abi3-win_amd64.whl (1.8 MB)
   ---------------------------------------- 1.8/1.8 MB 83.5 kB/s eta 0:00:00
Downloading yara_python-4.5.4-cp312-cp312-win_amd64.whl (1.8 MB)
   ---------------------------------------- 1.8/1.8 MB 132.0 kB/s eta 0:00:00
```

Figure 1. Installing Volatility 3 and its full dependency set via pip.

5.2 Acquiring the Memory Image with DumpIt

DumpIt was run directly on the target system and, when prompted, confirmed with 'y' to begin capture. It wrote the full physical memory contents to a timestamped .dmp file, alongside a small companion .json metadata file.

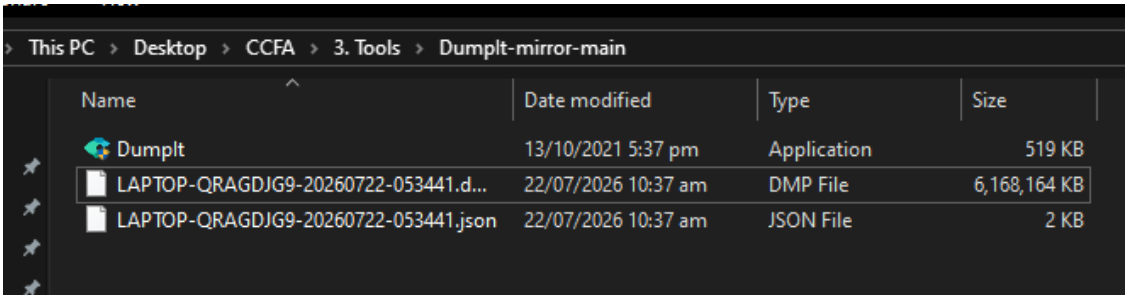


Figure 2. DumpIt's output folder: the captured memory image (LAPTOP-QRAGDIG9-20260722-053441.dmp, ≈ 5.9 GB) and its .json metadata file.

Finding: A complete physical memory image (≈ 6,168,164 KB) was captured successfully and used as the input file for every Volatility command that follows.

5.3 Identifying Available Windows Plugins

Volatility 3 organizes its plugins by operating system (windows.*, linux.*, mac.*). Running the tool with just the bare category name windows returns a disambiguation error listing every plugin available under that category, this is what produced the full plugin listing below, which serves as a reference for the operating-system-level artifacts Volatility 3 can extract from a Windows memory image.

Figure 3. Volatility 3's plugin disambiguation error, listing all available windows. plugins (e.g. windows.pslist, windows.netscan, windows.malware.malfind, windows.registry.hashdump) as a category reference.*

Of this full plugin catalog, the seven plugins used for this exercise are detailed in Sections 5.4–5.10 below, each with its purpose and findings.

5.4 windows.info: System Identification

Purpose: Confirms which build of Windows produced the memory image, the kernel base address, and the symbol table Volatility resolved against, establishing that the correct OS profile was used before trusting any later plugin's output.

```
python vol.py -f "...\LAPTOP-QRAGDIG9-20260722-053441.dmp" windows.info
```

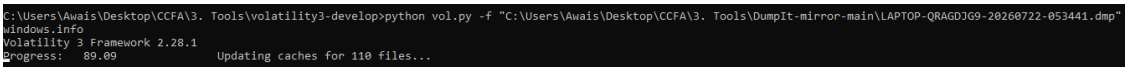


Figure 4. windows.info command invocation.

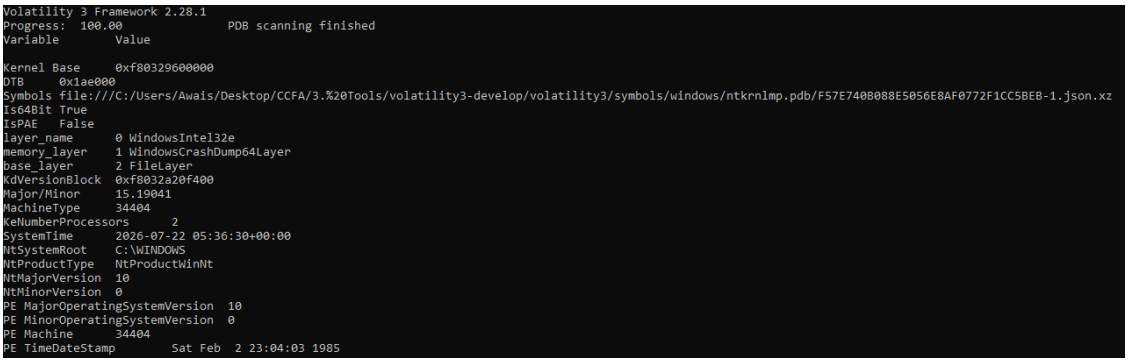
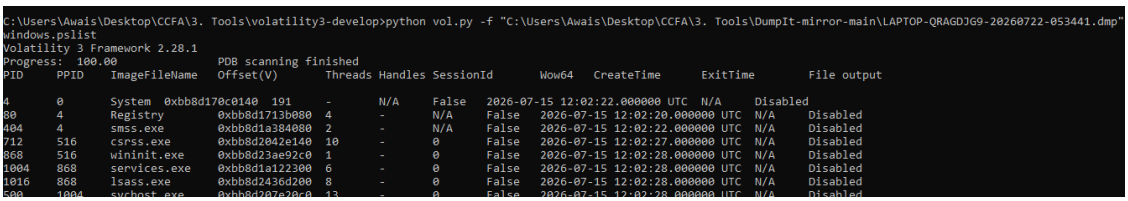


Figure 5. windows.info output: kernel base, symbol file, and system version details.

Finding: Windows 10 (NtMajorVersion 10), 2 logical processors, 64-bit, SystemTime 2026-07-22 05:36:30 UTC, confirming the image is a valid, correctly-profiled Windows 10 memory capture matching the capture time in Section 5.2.

5.5 windows.pslist: Running Process List

Purpose: Enumerates every active process at the moment of capture by walking the kernel's process list, giving PID, parent PID, thread/handle counts, and creation time for each, the starting point for almost any memory investigation.



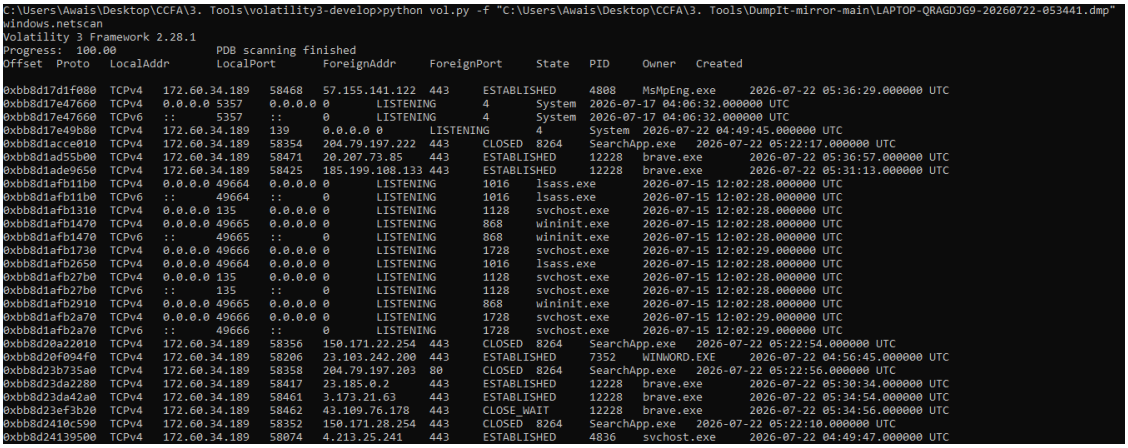
```
C:\Users\Awais\Desktop\CCFA\3. Tools\volatility3-develop>python vol.py -f "C:\Users\Awais\Desktop\CCFA\3. Tools\DumpIt-mirror-main\LAPTOP-QRAGDJG9-20260722-053441.dmp"
windows.pslist
Volatility 3 Framework 2.28.1
Progress:  100.00               PDB scanning finished
PID     PPID    ImageFileName   Offset(V)       Threads Handles SessionId       Wow64   CreateTime      ExitTime        File output

4       0       System  0xbb8d170c0140  191     -       N/A     False   2026-07-15 12:02:22.000000 UTC  N/A     Disabled
80      4       Registry        0xbb8d1713b080  4       -       N/A     False   2026-07-15 12:02:20.000000 UTC  N/A     Disabled
404     4       smss.exe        0xbb8d1a384080  2       -       N/A     False   2026-07-15 12:02:22.000000 UTC  N/A     Disabled
712     516     csrss.exe       0xbb8d2042e140  10      -       0       False   2026-07-15 12:02:27.000000 UTC  N/A     Disabled
868     516     wininit.exe     0xbb8d23ae92c0  1       -       0       False   2026-07-15 12:02:28.000000 UTC  N/A     Disabled
1004    868     services.exe    0xbb8d1a122300  6       -       0       False   2026-07-15 12:02:28.000000 UTC  N/A     Disabled
1016    868     lsass.exe       0xbb8d2436d200  8       -       0       False   2026-07-15 12:02:28.000000 UTC  N/A     Disabled
500     1004    svchost.exe     0xbb8d207e20c0  13      -       0       False   2026-07-15 12:02:28.000000 UTC  N/A     Disabled
```

Figure 6. windows.pslist output: System, Registry, smss.exe, csrss.exe, wininit.exe, services.exe, lsass.exe, and further processes with their PIDs and creation times.

Finding: A normal, expected core Windows process set was present (System, Registry, smss.exe, csrss.exe, wininit.exe, services.exe, lsass.exe), each with a plausible parent PID and no unexpected process names, no findings of concern at this stage.

5.6 windows.netscan: Network Connections

Purpose: Scans memory for network-connection structures (TCP/UDP endpoints), independent of the live pslist walk, recovering both active and some recently-closed connections along with the owning process, which is useful for identifying command-and-control traffic, data exfiltration, or unexpected listening services.



```
C:\Users\Awais\Desktop\CCFA\3. Tools\volatility3-develop>python vol.py -f "C:\Users\Awais\Desktop\CCFA\3. Tools\DumpIt-mirror-main\LAPTOP-QRAGDJG9-20260722-053441.dmp"
windows.netscan
Volatility 3 Framework 2.28.1
Progress:  100.00               PDB scanning finished
Offset  Proto   LocalAddr       LocalPort       ForeignAddr     ForeignPort     State   PID     Owner   Created

0xbb8d17d1f080  TCPv4   172.60.34.189   58468   57.155.141.122  443     ESTABLISHED     4808    MsMpEng.exe     2026-07-22 05:36:29.000000 UTC
0xbb8d17e47660  TCPv4   0.0.0.0 5357    0.0.0.0 0       LISTENING       4       System  2026-07-17 04:06:32.000000 UTC
0xbb8d17e47660  TCPv6   ::      5357    ::      0       LISTENING       4       System  2026-07-17 04:06:32.000000 UTC
0xbb8d17e49b80  TCPv4   172.60.34.189   139     0.0.0.0 0       LISTENING       4       System  2026-07-22 04:49:45.000000 UTC
0xbb8d1acce010  TCPv4   172.60.34.189   58354   204.79.197.222  443     CLOSED  8264    SearchApp.exe   2026-07-22 05:22:17.000000 UTC
0xbb8d1ad55b00  TCPv4   172.60.34.189   58471   20.207.73.85    443     ESTABLISHED     12228   brave.exe       2026-07-22 05:36:57.000000 UTC
0xbb8d1ade9650  TCPv4   172.60.34.189   58425   185.199.108.133 443     ESTABLISHED     12228   brave.exe       2026-07-22 05:31:13.000000 UTC
0xbb8d1afb11b0  TCPv4   0.0.0.0 49664   0.0.0.0 0       LISTENING       1016    lsass.exe       2026-07-15 12:02:28.000000 UTC
0xbb8d1afb11b0  TCPv6   ::      49664   ::      0       LISTENING       1016    lsass.exe       2026-07-15 12:02:28.000000 UTC
0xbb8d1afb1310  TCPv4   0.0.0.0 135     0.0.0.0 0       LISTENING       1128    svchost.exe     2026-07-15 12:02:28.000000 UTC
0xbb8d1afb1470  TCPv4   0.0.0.0 49665   0.0.0.0 0       LISTENING       868     wininit.exe     2026-07-15 12:02:28.000000 UTC
0xbb8d1afb1470  TCPv6   ::      49665   ::      0       LISTENING       868     wininit.exe     2026-07-15 12:02:28.000000 UTC
0xbb8d1afb1730  TCPv4   0.0.0.0 49666   0.0.0.0 0       LISTENING       1728    svchost.exe     2026-07-15 12:02:29.000000 UTC
0xbb8d1afb2650  TCPv4   0.0.0.0 49664   0.0.0.0 0       LISTENING       1016    lsass.exe       2026-07-15 12:02:28.000000 UTC
0xbb8d1afb27b0  TCPv4   0.0.0.0 135     0.0.0.0 0       LISTENING       1128    svchost.exe     2026-07-15 12:02:28.000000 UTC
0xbb8d1afb27b0  TCPv6   ::      135     ::      0       LISTENING       1128    svchost.exe     2026-07-15 12:02:28.000000 UTC
0xbb8d1afb2910  TCPv4   0.0.0.0 49665   0.0.0.0 0       LISTENING       868     wininit.exe     2026-07-15 12:02:28.000000 UTC
0xbb8d1afb2a70  TCPv4   0.0.0.0 49666   0.0.0.0 0       LISTENING       1728    svchost.exe     2026-07-15 12:02:29.000000 UTC
0xbb8d1afb2a70  TCPv6   ::      49666   ::      0       LISTENING       1728    svchost.exe     2026-07-15 12:02:29.000000 UTC
0xbb8d20a22010  TCPv4   172.60.34.189   58356   150.171.22.254  443     CLOSED  8264    SearchApp.exe   2026-07-22 05:22:54.000000 UTC
0xbb8d20f094f0  TCPv4   172.60.34.189   58206   23.103.242.200  443     ESTABLISHED     7352    WINWORD.EXE     2026-07-22 04:56:45.000000 UTC
0xbb8d23b735a0  TCPv4   172.60.34.189   58358   204.79.197.203  80      CLOSED  8264    SearchApp.exe   2026-07-22 05:22:56.000000 UTC
0xbb8d23da2280  TCPv4   172.60.34.189   58417   23.185.0.2      443     ESTABLISHED     12228   brave.exe       2026-07-22 05:30:34.000000 UTC
0xbb8d23da42a0  TCPv4   172.60.34.189   58461   3.173.21.63     443     ESTABLISHED     12228   brave.exe       2026-07-22 05:34:54.000000 UTC
0xbb8d23ef3b20  TCPv4   172.60.34.189   58462   43.109.76.178   443     CLOSE_WAIT      12228   brave.exe       2026-07-22 05:34:56.000000 UTC
0xbb8d2410c590  TCPv4   172.60.34.189   58352   150.171.28.254  443     CLOSED  8264    SearchApp.exe   2026-07-22 05:22:10.000000 UTC
0xbb8d24139500  TCPv4   172.60.34.189   58074   4.213.25.241    443     ESTABLISHED     4836    svchost.exe     2026-07-22 04:49:47.000000 UTC
```

Figure 7. windows.netscan output: TCP/IPv4 and IPv6 endpoints with local/foreign address, port, state, and owning process.

Finding: All identified connections resolved to expected, legitimate processes: MsMpEng.exe (Windows Defender, ESTABLISHED to a Microsoft telemetry/definition-update endpoint), brave.exe (multiple ESTABLISHED HTTPS connections, consistent with normal browsing), SearchApp.exe (Windows Search, CLOSED connections to Microsoft/Bing endpoints), and svchost.exe/lsass.exe/wininit.exe listening on standard Windows service ports (135, 139, 49664–49666). No connections to unrecognized or suspicious foreign addresses were observed.

5.7 windows.pstree: Process Hierarchy

Purpose: Presents the same process list as pslist but organized by parent–child relationship (indentation shows which process launched which), which is useful for spotting a process running from an unexpected parent, a classic sign of process injection or masquerading.

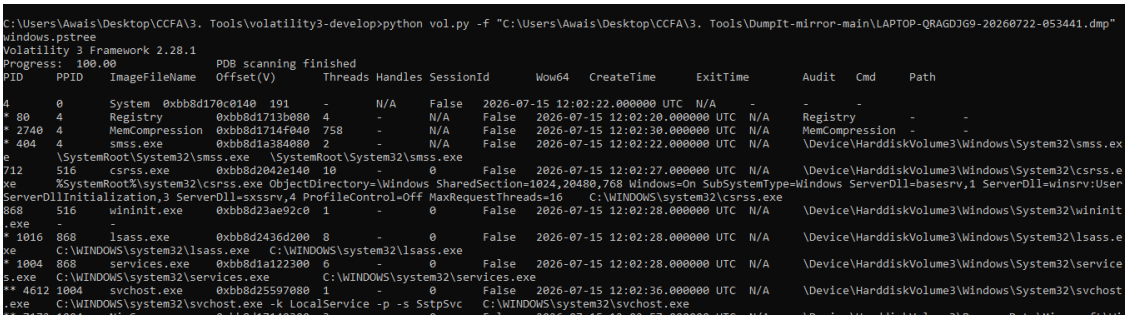


Figure 8. windows.pstree output showing System → smss.exe / Registry / MemCompression, and csrss.exe / wininit.exe → lsass.exe, services.exe → svchost.exe, with full paths and command lines.

Finding: Every process observed had a parent consistent with the normal Windows Vista/10 boot sequence (System spawning smss.exe and Registry; wininit.exe spawning services.exe and lsass.exe; services.exe spawning svchost.exe instances), no orphaned or unexpectedly-parented processes were found.

5.8 windows.cmdline: Process Command-Line Arguments

Purpose: Recovers the exact command line each process was launched with, including service-hosting arguments passed to generic executables like svchost.exe, which is critical for distinguishing a legitimate system service from a malicious process disguising itself under a common process name.

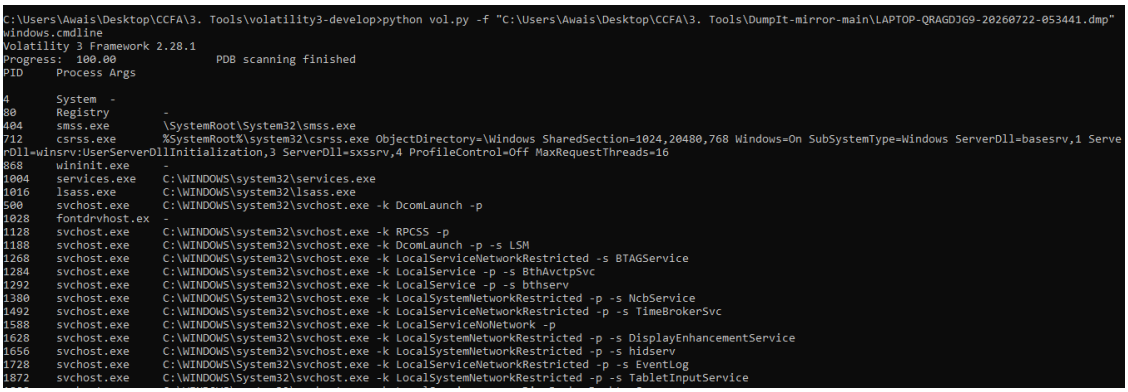


Figure 9. windows.cmdline output listing each process's full command line, including svchost.exe instances with their -k service-group arguments (e.g. -k LocalService -p -s BthAvctpSvc).

Finding: Every svchost.exe instance carried a plausible, named -k service-group argument (DcomLaunch, RPCSS, LocalServiceNetworkRestricted, EventLog, and others) matching known legitimate Windows service groups, none were found running with a blank, malformed, or suspicious argument, which would be the classic red flag for a process masquerading as svchost.exe.

5.9 windows.malware.malfind: Injected Code Detection

Purpose: Scans each process's memory for regions that are both writable and executable (PAGE_EXECUTE_READWRITE) with no backing file on disk, which is a strong indicator of injected shellcode or a reflectively-loaded malicious module, since legitimate code is normally either executable-only (loaded from a file) or writable-only (ordinary data), rarely both at once.

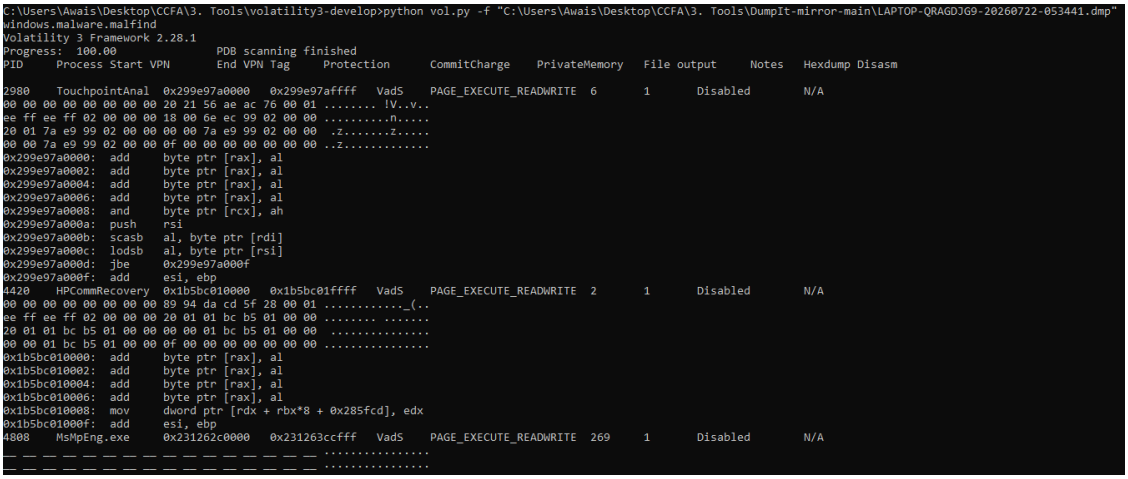


Figure 10. windows.malware.malfind output flagging three processes with PAGE_EXECUTE_READWRITE regions: TouchpointAnalytics (PID 2988), HPCommRecovery (PID 4420), and MsMpEng.exe (PID 4808), with hex dumps and disassembly.

Finding: Three processes were flagged: TouchpointAnalytics and HPCommRecovery (both pre-installed HP laptop utilities) and MsMpEng.exe (Windows Defender's own scanning engine).

> **TECHNICAL NOTE: ASSESSING THE MALFIND FLAGS**
>
> malfind is intentionally broad, it flags the memory permission pattern associated with code injection, not injection itself, so a certain rate of false positives on a clean system is expected and must be triaged rather than reported at face value.
>
> MsMpEng.exe is a textbook example: antivirus engines legitimately allocate RWX memory for their own emulation/unpacking sandboxes to safely detonate suspicious code, which is exactly the pattern malfind is designed to catch. Flagging Windows Defender's own process is one of the most common and well-documented malfind false positives.
>
> TouchpointAnalytics and HPCommRecovery are OEM telemetry/support utilities bundled with HP laptops (consistent with the HP hardware identified in the companion Windows Registry examination of this portfolio); vendor utilities of this kind sometimes use RWX regions for legitimate JIT-compiled or self-updating code, another common source of benign flags.
>
> Assessment: all three flags are consistent with expected false positives on a clean baseline system, not evidence of compromise. In a real investigation, the next step for any of these would be to export the flagged region (--dump), hash it, and check it against a known-good hash or submit it for reputation/YARA analysis rather than accept or dismiss the flag on process name alone.

5.10 windows.filescan: Open File Handles

Purpose: Scans memory for file-object structures, recovering the names of files that were open (or recently open) at capture time, including system files, NTFS metadata structures, and driver files, which is useful for identifying files a suspicious process had open that may not otherwise be visible from a live directory listing.

```
C:\Users\Awais\Desktop\CCFA\3. Tools\volatility3-develop>python vol.py -f "C:\Users\Awais\Desktop\CCFA\3. Tools\DumpIt-mirror-main\LAPTOP-QRAGDJG9-20260722-053441.dmp"
windows.filescan
Volatility 3 Framework 2.28.1
Progress:  100.00               PDB scanning finished
Offset  Name

0xbb8d1787c210  \Windows\System32\drivers\npfs.sys
0xbb8d1787c380  \Windows\System32\drivers\netbt.sys
0xbb8d1787c660  \Windows\System32\drivers\tdx.sys
0xbb8d1787cab0  \$Directory
0xbb8d1787cc20  \$Directory
0xbb8d1787d070  \Windows\System32\drivers\msfs.sys
0xbb8d1787d1e0  \Windows\System32\DriverStore\FileRepository\basicdisplay.inf_amd64_19e58b6267591a82\BasicDisplay.sys
0xbb8d1787d4c0  \Windows\System32\drivers\cimfs.sys
0xbb8d1787d910  \Windows\System32\DriverStore\FileRepository\basicrender.inf_amd64_eebb5a8467eb88ec\BasicRender.sys
0xbb8d1787dd60  \Windows\System32\drivers\tdi.sys
0xbb8d1788f210  \Windows\System32\drivers\pacer.sys
0xbb8d1788f4f0  \Windows\System32\drivers\netbios.sys
0xbb8d1788f7d0  \Windows\System32\drivers\afd.sys
0xbb8d1788f940  \Windows\System32\drivers\ndiscap.sys
0xbb8d1788fc20  \Windows\System32\drivers\afunix.sys
0xbb8d17890350  \Windows\System32\drivers\Vid.sys
0xbb8d17890ed0  \Windows\System32\drivers\vwififlt.sys
0xbb8d17989380  \$MftMirr
0xbb8d179894f0  \$LogFile
0xbb8d17989660  \$MapAttributeValue
0xbb8d179897d0  \$Secure:$SII:$INDEX_ALLOCATION
0xbb8d17989d90  \$Mft
0xbb8d1798a070  \$Extend:$I30:$INDEX_ALLOCATION
0xbb8d1798a350  \$Secure:$SDS:$DATA
```

Figure 11. windows.filescan output listing open file objects: Windows driver files (npfs.sys, netbt.sys, tdx.sys, afd.sys) and NTFS metadata structures ($LogFile, $Secure, $Extend).

Finding: The recovered file handles were entirely standard Windows kernel driver files and NTFS metadata structures (e.g. $LogFile, $Secure:$SDS:$DATA), consistent with normal system operation and containing no unexpected or user-suspicious file paths.

6. Narrowing an Investigation: Targeted Filtering

The plugins above were run unfiltered, returning every process/connection/region system-wide, appropriate for an initial baseline triage. Once a specific process becomes a focus of interest (for example, one of the malfind hits in Section 5.9, in a case where it could not be dismissed as a false positive), Volatility supports narrowing any plugin's scope directly, rather than manually filtering its full output:

- --pid, restrict the plugin to a single specific process ID.
- --name, filter by process name, supporting wildcard patterns.
- --regex, filter using a full regular expression for more complex pattern matching.
- --dump, combined with malfind, exports the flagged memory region itself to disk for hashing, YARA scanning, or manual reverse-engineering, rather than only reporting its presence.

For example, to focus malfind on a single suspicious PID:

```
# Target a specific suspicious PID vol -f memory.dmp windows.malware.malfind --pid
1724
```

To filter by process name pattern instead of a fixed PID:

```
# Filter by process name pattern vol -f memory.dmp windows.malware.malfind --name
"powershell*"
```

Combining a targeted PID filter with --dump extracts the flagged region itself for offline analysis:

```
# Use regex to find injected code in specific processes vol -f memory.dmp
windows.malware.malfind --pid 1724 --dump
```

Beyond these, Volatility also supports narrowing by memory region using --base, or combining several of the filters above at once, to isolate only the most relevant artifacts for a given threat model rather than paging through a full system-wide dump on every plugin run.

7. Findings Summary and Conclusion

> **TECHNICAL NOTE: OVERALL ASSESSMENT**

> System identification (windows.info) confirmed a valid Windows 10 image with a correctly resolved symbol table.
>
> Process listing and hierarchy (windows.pslist, windows.pstree) showed an expected, normally-parented Windows process set with no orphaned or misplaced processes.
>
> Network connections (windows.netscan) resolved entirely to legitimate, identifiable applications with no unrecognized foreign endpoints.
>
> Command-line arguments (windows.cmdline) showed every svchost.exe instance carrying a valid, named service-group argument.
>
> Injected-code detection (windows.malware.malfind) flagged three processes; all three were assessed as expected false positives (an antivirus engine and two OEM utilities) rather than evidence of compromise.
>
> Open file handles (windows.filescan) showed only standard system driver and NTFS metadata files.

This exercise reproduces the standard first-pass triage workflow an examiner runs against any newly acquired Windows memory image: confirm the image and profile, enumerate processes and their relationships, check network activity, and screen for injected code, before deciding whether deeper analysis is warranted. In this case, every plugin's output was consistent with a normal, uncompromised baseline system, but the same seven-plugin sequence, together with the targeted --pid/--name/--regex/--dump filtering shown in Section 6, is exactly what would be applied first against a genuinely suspicious system, where the findings at each step would be expected to look materially different.